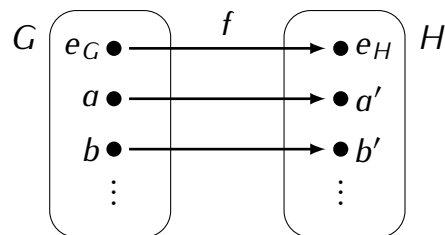


Definition 12.1

An *isomorphism of groups* is a group homomorphism which is both onto and 1-1.

**Theorem 12.2**

If $g: G \rightarrow H$ is an isomorphism then the inverse function $f^{-1}: H \rightarrow G$ is also an isomorphism.

Theorem 12.3

A homomorphism of groups $f: G \rightarrow H$ is an isomorphism if and only if $\text{Im}(f) = H$ and $\text{Ker}(f) = \{e\}$.

Definition 12.4

We say the group G is *isomorphic* to a group H if there exists an isomorphism $f: G \rightarrow H$. Then we write $G \cong H$.

Theorem 12.5

Isomorphism of groups is an equivalence relation:

- 1) For any group G we have $G \cong G$.
- 2) If G, H are groups such that $G \cong H$ then $H \cong G$.
- 3) If G, H, K are groups such that $G \cong H$ and $H \cong K$, then $G \cong K$.

Theorem 12.6 (Cayley's Theorem)

Let G be a finite group of order n . Then G is isomorphic to a subgroup of the symmetric group S_n .

Definition 12.7

An *automorphism* of a group G is an isomorphism $f: G \rightarrow G$.

Definition 12.8

Let G be a group. The *group of automorphisms* of G is the group $\text{Aut}(G)$ whose elements are automorphisms of G and the group operation is given by composition of automorphisms.